

Tajveer Singh Dhesi

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Work authorization: F-1 student visa (eligible for CPT / OPT)

EDUCATION

University of Chicago — *Ph.D., Computer Science* September 2025 – Present

- Advised by Prof. Nick Feamster (NOISE Lab)
- Research areas: web measurement, AI governance, online safety policy, Agentic AI Privacy

Colgate University — *B.A., Computer Science and Mathematics (double major)* August 2021 – May 2025

- Graduated *summa cum laude*, with High Honors in Computer Science
- Alumni Memorial Scholar (awarded to ~1% of class for excellence in research)
- Dean's Award with Distinction, 5 consecutive semesters
- Relevant coursework: Probability, Combinatorial Problem Solving, Data Analysis, Applied Machine Learning, Algorithmic Game Theory, Honors Thesis

University of Wollongong, Australia — *Visiting Student* February 2024 – June 2024

- Coursework in Data Mining & Knowledge Discovery, and Modern Artificial Intelligence

LANGUAGES & TECHNOLOGIES

Programming: Python (Selenium, requests, Playwright), R, Java, JavaScript, C, VB.NET, SQL, VBA

ML / AI: OpenAI API, Claude Console, Hugging Face transformers, local LLM inference (Ollama, mlx.lm), Privacy aware prompt engineering, evaluation pipeline design

Systems & tooling: Linux/Unix, Docker, Git, LaTeX, headless Chromium, MCP (Model Context Protocol)

Web: HTML, CSS, React.js, React Native, Cloud Firestore

RESEARCH EXPERIENCE

Graduate Researcher — **NOISE Lab, University of Chicago** September 2025 – Present

Advisor: Prof. Nick Feamster

Agentic AI Data leakage March 2026 – Present

- Designing a benchmark suite to measure **data leakage in agentic LLM systems** across realistic tasks (drafting emails, social-media posts, and similar information-relay workflows). Tasks contain mixed relevant and irrelevant sensitive information; the benchmark quantifies how often agents leak the irrelevant content despite being instructed to use only the relevant subset.
- Targeting both open-source (7B-32B parameter) and frontier closed-source agents, enabling direct utility-vs-leakage comparisons across deployment tiers relevant to enterprise and consumer use.
- Will propose and evaluate candidate defenses against the benchmark, characterizing the **utility-to-defense-efficacy trade-off** — directly relevant to NIST AI RMF guidance and emerging FTC and state-AG enforcement around AI privacy harms.
- Contributing to lab-wide work auditing content-moderation policy enforcement across major platforms, measuring gaps between stated guidelines and observed practice.

Transformer-Based Web Scraping for Age Gate Detection September 2025 – Present

- Extending site-arrival age-gate measurement (prior thesis work) to detect age gates at deeper purchase-flow interaction points (cart, checkout, account creation) using a transformer-based scraping agent that navigates without site-specific heuristics.
- Motivated by the limitations of single-page screenshot pipelines and increasing legislative attention to mid-flow age verification (UK LCCP 2025, NSW/Queensland delivery rules).

Senior Thesis Research — Colgate University

August 2024 – May 2025

"Measuring the Variety of Website Arrival Age Gates in the E-Cigarette, Alcohol and Gambling Industry"

- Built an automated measurement pipeline (Selenium + headless Chromium + GPT-4o-mini classifiers) to identify and classify site-arrival age gates across **4,985 international websites** in three regulated industries.
- Cataloged **595 age gates** across 8 country-code TLDs; found that virtually all rely on trivial self-attestation.
- Achieved **95.5% accuracy** on screenshot-based gate detection vs. 78.0% for HTML-only — empirically justifying vision-based classification for poorly-labeled DOM elements.
- Translated findings into policy framing: tied prevalence patterns to specific jurisdictional regimes (Canada's Vaping Promotion Regulations, UK Gambling Commission LCCP, NSW alcohol delivery rules).

Research Assistant — Computer Science Dept., Colgate University

May 2023 – September 2023

- Generated runtime and accuracy data for a deep-learning model (gRNAde) predicting RNA base sequences from tertiary PDB structures, using Weights & Biases for experiment tracking on university compute nodes.
- Co-authored the resulting preprint demonstrating that ligand-bound RNA structure learning diverges from unbound learning.

PUBLICATIONS

- T. S. Dhesi, N. Apthorpe, A. Gember-Jacobson. "Measuring the Prevalence and Variety of Online Age-Gates." Workshop paper, 2025. <https://conpro25.ieee-security.org/papers/dhesi-conpro25.pdf>
- T. S. Dhesi, A. W. Bannister, A. Manzourolajdad. "Deep Learning of Ligand-bound RNA Tertiary Structures Diverges from Learning Unbound Ones: A Case Study Using The gRNAde Software." bioRxiv, 2023. [doi:10.1101/2023.09.13.557627](https://doi.org/10.1101/2023.09.13.557627)

TEACHING EXPERIENCE

Teaching Assistant, University of Chicago, Department of Computer Science

September 2025 – Present

- *Introduction to Computer Security* (current) — Grading and student-facing support on system-level exploits (TOCTOU race conditions), RSA padding-oracle attacks, and web-scraping-based attacks.
- *Machine Learning for Computer Systems* — Graded a Netflix packet-trace analysis assignment applying ML to network performance characterization.
- *Creating Interactive Systems with User-Centered Design* — Mentored 3 student groups through a quarter-long project building interactive systems for specific user populations, advising on HCI methodology and user research design.

Teaching Assistant, Colgate University

August 2023 – December 2023

- Supported 50+ students across two sections of Discrete Structures (COSC 290) and one section of Data Structures & Algorithms (COSC 202), both taught in Java.

SELECTED PROJECTS

Maze Generation Game (JavaFX) — University of Rochester

November 2023

- Built a procedural maze generator using Kruskal's algorithm in a team of three; led the algorithmic implementation - *2nd place, DandyHacks Hackathon*. [Repository](#)

Full-Stack Developer — Sloop Software Group, Colgate University

May 2023 – December 2023

- Built a student rideshare iOS/Android app (React Native + Cloud Firestore) handling real-time ride requests and customer data.

HONORS, AWARDS & CERTIFICATIONS

- **Intro to Model Context Protocol**, Anthropic (April 2026)
- **The Laura Sanchis Award for Excellence in Research**, Colgate University (2025)
- **Excel Developer** - With comprehensive Portfolio Management Dashboard Project, IACT (Jan 2022)
- **From Excel to Power BI**, Coursera (July 2024)